Qn	Suggested Answer
1	
(a)(i)	The *linear momentum* of a body is defined as the product of its mass and its velocity and it is in the direction of the velocity.
(a)(ii)	The total final momentum of a system after a collision is equal to total initial momentum of the system before the collision
	provided no net external force acts on the system.
(b)(i)	Applying the principle of conservation of momentum, $m_A v_A + m_B v_B = 0$ $2.0 v_A = -1.0 v_B$ $v_B = -2.0 v_A \quad \cdots (1)$
	Applying law of conservation of energy, $\frac{1}{2} m_A v_A{}^2 + \frac{1}{2} m_B v_B{}^2 = 12 \quad \cdots (2)$
	Substitute equation (1) into equation (2), $\frac{1}{2}(2.0) v_A{}^2 + \frac{1}{2}(1.0)(2.0 v_A)^2 = 12$ $v_A{}^2 + 2.0 v_A{}^2 = 12$ $\lvert v_A \rvert = 2.0 \text{ m s}^{-1}$
	$\lvert v_B \rvert = 4.0 \text{ m s}^{-1}$
(b)(ii)	A and B are moving towards each other. At any instant of time, the speed of B is twice that of A, distance covered by B is twice that of A when they collide.
	$x_A + x_B = 0.90$ $x_A + 2x_A = 0.90$
	$x_A = 0.30$ m
2	